Sthiti Padhi

Education

Expected June 2027 **Dual Degree (B.Tech. + M.Tech.) in Electrical Engineering**, *Indian Institute of Technology Bombay*, Mumbai, India
Minor from the Centre for Machine Intelligence and Data Science
- Advisor: Prof. Laxmeesha Somappa
- Cumulative Performance Index (CPI): 9.63/10.00

Aug 2025 – Dec 2025 **Semester Exchange (Autumn 2025), D-ITET**, *ETH Zurich*, Zurich, Switzerland
- Graded Performance (GPA): 4.85/6.0

Publications

2024 D. Christmann, J. F. Gutierrez, **S. Padhi**, P. Plörer, A. Takur, S. Silvestri, and A. Gomez, "Space Debris Removal using Nano-Satellites controlled by Low-Power Autonomous Agents," in *Proc. European Conference on Multi-Agent Systems (EUMAS)*, Demo Track, 2024 (peer-reviewed demonstration paper; arXiv:2601.00465).

Research Experience

Jan 2026 – Present **Master's Thesis: Autoencoder for Neural Signals**, *IIT Bombay*, Mumbai, India
Advisor: Prof. Laxmeesha Somappa | Supervised Research Exposition & Dual Degree Project
- Achieved >75% bandwidth reduction with <10% error by benchmarking **DCT**, **WHT** and **downsampling** on seizure vs. non-seizure iEEG recordings from the **SWEC-ETHZ** database (MATLAB)
- Designed a sparse **autoencoder** to improve baseline **Walsh-Hadamard Transform** (WHT) compression; scaled a single-layer design into a **two-stage autoencoder** that handles channel and temporal mixing separately
- Executed **quantization-aware training** (QAT) to enable low-power, low-compute on-chip hardware execution

Aug 2025 – Dec 2025 **Energy Efficient Ball Towards 360 Vision**, *ETH Zurich*, Switzerland
10-credit mobility project, Autumn 2025 | Guide: Dr. Phillipp Mayer
Integrating on-board person detection with image capture in a 360° multi-camera ball
- Trained **TinyissimoYOLOv8** on a curated person-class **COCO subset** (320×320 images, exported at 320×256)
- Applied **post-training quantization** to fit the model within the tight **GAP9** cluster **L2 memory**
- Integrated an end-to-end pipeline from image capture to person detection using dynamic **L1/L2 memory sharing**

May 2024 – Jul 2024 **Research Intern**, *Technische Universität Braunschweig*, Germany
On-site internship | Guide: Prof. Andres Gomez
- Developed **AgentSpeak** and embedded **Belief-Desire-Intention** (BDI) agent code for nanosatellites in a low-power multi-agent system
- Integrated agent functions on **OpenThread** using **CoAP** in **Zephyr RTOS** on **nRF boards**; co-authored the resulting demonstration accepted at **EUMAS 2024**
- Built a VSCode debugging platform on Linux by integrating **OpenOCD** and **Docker**, enabling debugging of **STM32L432** and CC26x2 boards on Ubuntu
- Explored the **OpenThread CLI** and enabled **CoAP** communication between two CC26x2 boards

Industry & Technical Experience

May 2025 – Jul 2025 **Embedded Software Engineering Intern**, *Google India*, Bengaluru
Little Kernel Operating System Enhancements
- Engineered a **thread-kill** feature in **Little Kernel** with resource cleanup and recovery for **mutexes**, **spinlocks** and **semaphores**
- Analyzed the **thread lifecycle management** and scheduler of Little Kernel, **Zephyr** and **Zircon**
- Integrated unique **thread identifiers** with system logs to enable contextual, thread-level diagnostic tracking

May 2026 – Jul 2026 **Software Engineering Intern (Google Cloud)**, *Google India*, Bengaluru

Evaluation framework for an alert system: estimating accuracy of a causal attribution model for user latency

- Implemented **unsupervised accuracy estimation** for **Bayesian Networks** as a baseline without ground-truth data
- Researched **Confidence-Based Performance Estimation** to quantify model performance under covariate shift
- Formulated **Expected Calibration Error** (ECE), **Adaptive ECE** and reliability curves for model calibration
- Developed and unit-tested a **CLI tool** and shared library to automate deployment of model-evaluation metrics

Jan 2023 – Feb 2025 **Electronics and Control Subsystem Lead**, *IIT Bombay Mars Rover Team*

Advisor: Prof. P.J. Guruprasad | Student initiative to build a prototype Mars Rover

- Ranked **6th** in the **International Rover Design Challenge** and **12th** in the **International Rover Challenge**
- Integrated **FPV cameras** and a **Mimosa C5c** PTP bridge, and explored **LoRa**, **LPWAN** and **MIMO** for the rover communication setup
- Implemented navigation with real-time **ArUco marker detection** and **Canny edge detection** using **OpenCV**

Selected Projects

Jan 2025 – Apr 2025 **Morphobot: Autotransforming Bot**, *Course Project, Electronic Design Lab*, Best-Project Award, YUGM 2025

- Built a hybrid ground-air robot inspired by Caltech's **Morphobot** around a single **SpeedyBee flight controller** coordinating DC, BLDC and servo systems: 3 min flight, 15 min drive, $<$4 s transformation
- Customized **ArduPilot firmware** for aerial (UAV) and ground (rover) state transitions, controlled through a standard UAV RC transmitter
- Validated with **hardware-in-the-loop** and **software-in-the-loop** (SITL) simulation in Mission Planner

Jan 2025 – Apr 2025 **Embedded Systems Design**, *Course Lab & Project*

- Developed Python-based game logic on the **ARM processing system** of a Zynq SoC with **analog joysticks** read via ADC and AXI GPIO
- Integrated an **OpenCV** pipeline with triple-buffered VDMA to output low-latency 800×600 graphics at 60 FPS
- Implemented AXI connect and the Zynq processor in Xilinx Vivado on a PYNQ-Z2 FPGA board; interfaced timers, LEDs and ADC on a TIVA-C using Code Composer Studio

Aug 2024 – Nov 2024 **Heart Rate Detection and Display on ESP32**, *Course Project, DH307 R&D Project*

- Established communication between an **ESP32** and a display, benchmarking **BLE**, **Bluetooth**, **Wi-Fi** and **ESP-NOW**
- Tested a heart-rate algorithm on a MAX30102 **PPG sensor** using raw **IR sensor values**
- Integrated the **LVGL GUI** library with an **ESP32-S3 TFT** display to render real-time heart-rate waveforms

Aug 2024 – Nov 2024 **CMOS Standard Cell & DWT Architecture**, *Course Project, EE671 VLSI Design*

- Implemented an 8×8 **Discrete Wavelet Transform** (DWT) on images in Verilog, followed by **RTL synthesis** and **static timing analysis**
- Designed a **3-input AND** gate, **inverter** and **negative-edge D flip-flop** using the Sky130 PDK in Magic
- Characterized **slack time**, **hold time**, and **static** and **dynamic power** of the CMOS-designed blocks

Apr 2024 – May 2024 **IITB-CPU Processor Design**, *Course Project, EE309 Microprocessors*

- Team of 4: designed a **16-bit RISC** CPU on **Harvard architecture**, optimized for **CPI** using 6-stage pipelined and multicycle implementations
- Implemented **hazard mitigation** using data forwarding and minimized data dependencies
- Designed the **datapath**, **controller** and memory system in Quartus using **VHDL**

Additional Coursework Projects

Aug 2025 – Dec 2025 **Image Analysis and Computer Vision**, *ETH Zurich*

Reconstructed a 3D world projection from dual perspective 2D images using camera calibration; implemented k-means and kNN image segmentation ($>$59% accuracy); designed a 2-layer CNN for 5-class classification and used transfer learning to expand to 10 classes

Oct 2024 – Nov 2024 **JPEG Image Compression**, *CS663*

Developed a JPEG system using the 2D DCT, quantization and Huffman coding; reviewed literature and applied a non-separable 2D-KLT instead to improve efficiency

Jul 2024 – Nov 2024 **Machine Learning & Feature Extraction**, *Introduction to Machine Learning*

Implemented **Logistic Regression, SVM,** and **MLP** from scratch in **NumPy**/**SciPy** with gradient descent and backpropagation, evaluated via **Accuracy, Precision, F1-Score**, and analyzed learned feature representations using **K-Means**/**DBSCAN** clustering

Aug 2024 – Nov 2024 **Hybrid Approach for MRI Image Segmentation**, *Wavelets*
Developed an MRI tissue segmentation framework combining **Multi-Level Wavelet CNNs** and **Self-Organizing Maps**, extracting 16 multi-resolution subbands via 2-level **DWT** on the IBSR dataset and clustering the wavelet feature vectors with an unsupervised **SOM** pipeline to classify brain tissues

Oct 2023 – Nov 2023 **Vibration Control in Chemical Plant using ML**, *Programming for Data Science*
Built **Multiple Linear Regression** models to quantify system criticality ($R^2 = 0.990$), reduced **multicollinearity** across **240** time-series parameters via **PCA**, and isolated the **30** most significant controllable parameters through **iterative p-value thresholding** ($R^2 = 0.905$)

Honors & Awards

2025 Best-Project Award and Special Mention, YUGM 2025 (Morphobot)

2025 Semester Exchange Selection (1 of 2 students), D-ITET, ETH Zurich

2022–26 Academic Proficiency Award in Psychology, Management and Optoelectronic Devices, IIT Bombay

2022 Joint Entrance Examination (JEE) Mains: All India Rank 959 (99.90 percentile); JEE Advanced: 99.07 percentile

2020 Kishore Vaigyanik Protsahan Yojana (KVPY) SA Scholarship, All India Rank 388 among 50,000+ candidates

Teaching Experience

Jul 2026 – Present **Teaching Assistant, Probability and Random Processes**, *Dept. of Electrical Engineering, IIT Bombay*
100+ second-year students

Jan 2025 – Apr 2025 **Teaching Assistant, Linear Algebra and Differential Equations**, *Dept. of Mathematics, IIT Bombay*
First-year students

Service

Aug 2026 – Present **Peer Mentor, Student Mentorship Program**, *IIT Bombay*
Mentored 2 undergraduate students on course planning, research opportunities and professional development

Jun 2025 – Jul 2025 **Volunteer Teaching Assistant, ELIT**
Mentored 4 students to improve spoken English through weekly conversation-practice sessions

May 2023 – Mar 2024 **Secretary, EE Alumni and Corporate Engagement Cell**, *IIT Bombay*
- Organized department tours and exhibitions for 70+ **IIT Bombay alumni**, facilitating **INR 30 million** (approx. USD 360K) in donations
- Led the **Alumni Networking Session** at the department fest Impulse and a Freshie Lab Tour; assisted in organizing the 61st Convocation of the EE Department

Technical Skills

Programming Python, Embedded C, C, C++, Assembly, VHDL, Verilog, SystemVerilog, Shell, LaTeX, SQL, AgentSpeak

Embedded nRF, STM32, CC26x2, ESP32, TIVA-C, GAP9, Zynq (PYNQ-Z2); Zephyr RTOS, OpenThread, CoAP, BLE, ESP-NOW

Tools MATLAB, Linux, Git, Fusion 360, Quartus, Vivado, Code Composer Studio, Keil IDE, LTSpice, Magic

Libraries ROS, ArduPilot, NumPy, OpenCV, SciPy, Scikit-learn, PyTorch, Pygame, Pandas

Coursework

- Embedded System Design
- VLSI Design
- VLSI-1: HDL Design based on FPGAs
- Wearable Technology
- Communication Networks
- Applied Linear Algebra
- Linear System Theory
- Markov Systems and Queueing Theory
- Introduction to Machine Learning
- Digital Image Processing
- Image Analysis and Computer Vision